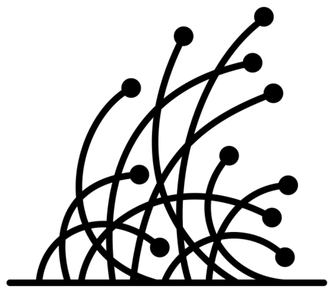


RNAseq

Considerations for fungi

Dr Johanna Rhodes

Learning outcomes

- **Familiarise terminology and file formats**
  - read, library, flowcell, lane, run, FASTQ, FASTA
- **Familiarise with basic commands**
  - gzip, gunzip, head
- **Reference-based RNAseq analysis**
  - theory
  - practice - splice-aware approaches
- ***De novo* transcriptomics**
  - theory, software used

Terminology

A refresher

- **Read**: a single sequence produced from a sequencer
  - e.g. a sequencing machine *reads* a molecule, and this is what it thinks it is
- **Library**: a collection of DNA fragment that have been prepared for sequencing. This is generally talking about individual samples
- **Flowcell**: a chip on which DNA is loaded and provided to the sequencer
- **Lane**: one portion of a flowcell
  - used for technical replicates or different samples
- **Run**: an entire sequencing reaction from start to finish

Data

Downloadable :)

- Data for today is publicly available from PRJEB57846
  - Download one set of paired-end reads each
- Other data will be made available via Github

File formats

FASTQ

- Most widely used format
- Contains more information that FASTA
- Each sequence requires four lines:
    1. Starts with ‘@’ and contains sequence identifier and description
    2. Sequence
    3. Starts with ‘+’ and can contain information
    4. Quality scores
- Compressed with ‘gzip’ - .gz extension
    - Decompress with ‘gunzip’
    - `gunzip -c filename.fastq.gz > file1.fastq`
    - `head file1.fastq`

```
@VH00344:2:AAAKMY2M5:1:1101:64074:1246 1:N:0:AGCACG
GCTGGATTATGGCGGCACGCCTCTACGGCAGCAGCCAGGCTNNACGCNCCGCCNNNNNNNGNCCNNNCNGGNNGNNNGGNNGCGCCGACGCCGCGCCGGCGC
GGTNNTGGCGGNGNCCCGNCNCNGGGGCCTGN
+
CCCCCCCCCCC-;-CCCCC;CCC;C;CCCCC-CCCCC-C;-##CC;;#C;C;C#######C#;C###C#CC##C###CC##C-C-;CC;C----;-CCC-C-
CCC##CCC;CC#C#CCCC#;#C#CCCCCCC;#
@VH00344:2:AAAKMY2M5:1:1101:64453:1246 1:N:0:AGCACG
AAAGGATCCATCCCCGCCCGTGTGACCGTGCCAGGCGTGGCNCCCAGNCCCCCNNNNNNNGNCCCNNCNCANCCNNNCGNNGCGGGGTCCGCGCCGGCGCGG
GCANNAGGCGCNGNCCCCNCNGNGCGCGCTC
+
CC-CCCCCCCCC-CCCCCCCCCCC;;-CCCCCCCC;CCCCC#-CCCC#CC-CC#######C#CC;##C#;-#-C###CC##CCCCCCCCCCCCC;CC-C;C;
CCC##CCCCCC#C#;CCC#C#C#C;C;C-CC
@VH00344:2:AAAKMY2M5:1:1101:65097:1246 1:N:0:ATCACG
CGACTTCCCTTAGCTCCATGGTTCTCGCAACGAGCGGCGGTNCACCGGGGAGCNNNNNGGCNGGGCNCNGTNCGNNNGGNNGCGCCAACCGGCCTGCCGGCG
CGTNNGCGCCCNGNACGCNGCCNCCCCCCTGN
```

Basic commands

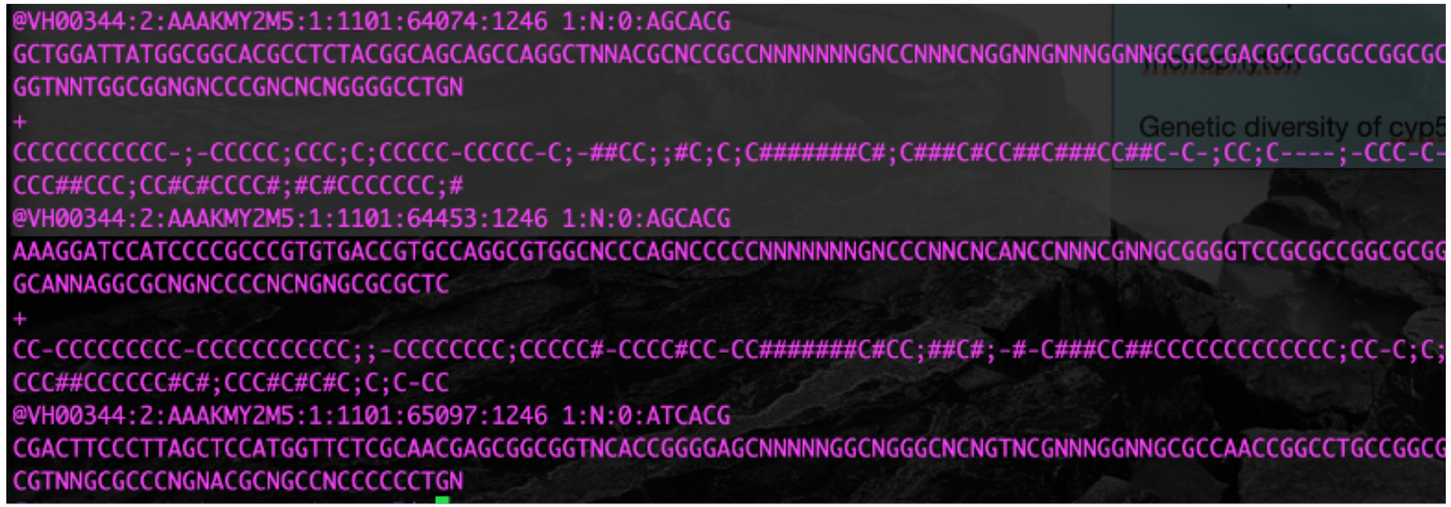
Reminder

- ‘gunzip’ decompresses the file
  - adding ‘-c’ makes a copy so the original file stays compressed, and a new, decompressed file is made
    - Better - saves time and prevents corruption of files (great power comes great responsibility)
- ‘head’ command prints lines from the top of the file
  - you can specify how many lines e.g.
  - `head -4 file1.fastq`
- ‘tail’ shows the lines at the bottom of the file

File formats

FASTQ

- Format is similar to DNAseq FASTQ in terms of content
- Casava 1.8 is used in Illumina 1.9 encoding of FASTQ files (Phred+33)
- Therefore, ‘!’ is lowest quality
- ‘~’ is highest quality
- Sequence reads tend to have lower quality due to errors at the end than at the start



File formats

FASTA

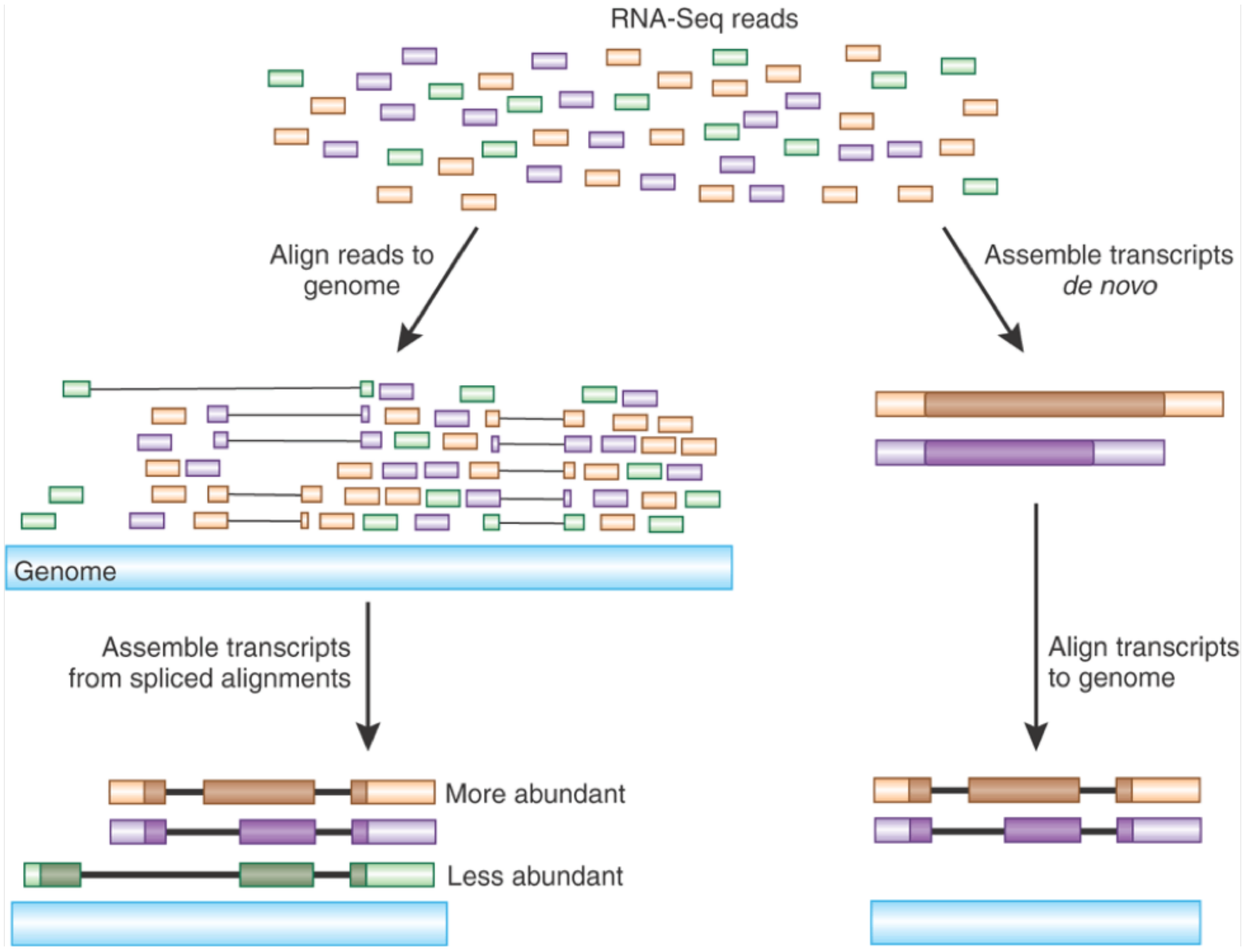
- Always starts with “>” followed by an identifier. The rest of the line (a description) is optional.
- Nucleotide or animo acid sequence follows on subsequent lines
- This format is used for reference genomes, and reference-based alignment workflows
    - Output from *de novo* alignments

```
@login-a live]$ head -5 GCF_000002655.1_ASM265v1_genom
4.1 Aspergillus fumigatus Af293 chromosome 1, whole ge
aaccctaaccctaaccctaaccctaaccctaaccctaaccctaaccctaaccct
ccctaaccctaaccctaaccctaaccctaaccctaacccttTAGGCAACTGCAGCTTCAA
GCTAGGTTTCCTGGTTCTTGGAACGACATTGTTCTCACCTAGTGTGATGAGCGT
CTTTGATCGAGTTTTCAGCGTTCAAGGAGGTCCCGTGTATGCTGTTCTGCGTAG
@login-a live]$
```

RNAseq

Strategies and theory

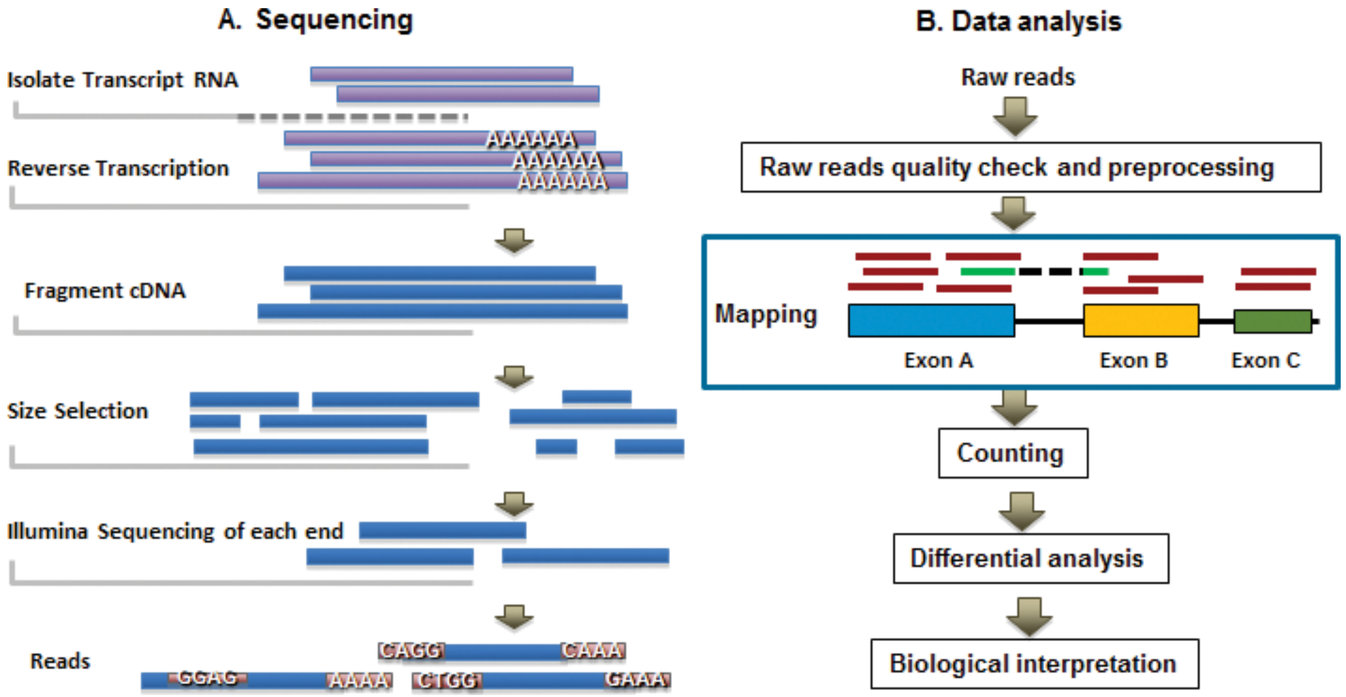
- Identify differentially expressed genes (DEGs)
- Identify co-regulated genes
- Time-series —> networks
- Annotation of genome
- Alternative splicing



Alignment-based transcriptomics

Strategy

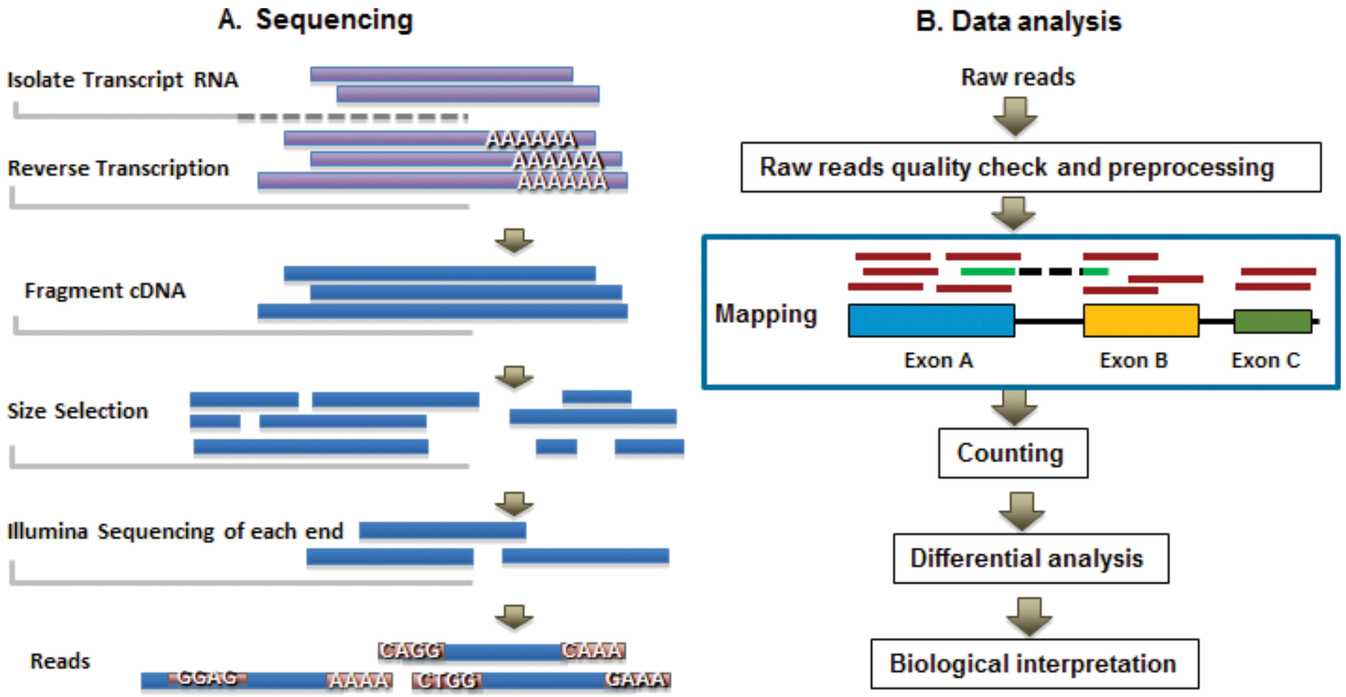
- If species of interest has been sequenced before, there is a strong possibility that a reference genome is available
- If there is not a reference genome available, then a *de novo* assembly approach needs to be employed



Alignment-based transcriptomics

Replicates

- Replicates are essential to correct for variation due to differences within/among organisms, cells, sequencing libraries/ machines etc.
- Technical replicates
  - e.g. different library preps from the same RNA sample
  - Account for batch effects e.g. PCR amplification
- Biological replicates
  - RNA from independent cells/tissues



De novo transcriptomics

Trinity

- Trinity is a software package used to perform *de novo* transcriptomic assembly
  - used when there is no reference genome available
  - looking for novel genes

De novo transcriptomics

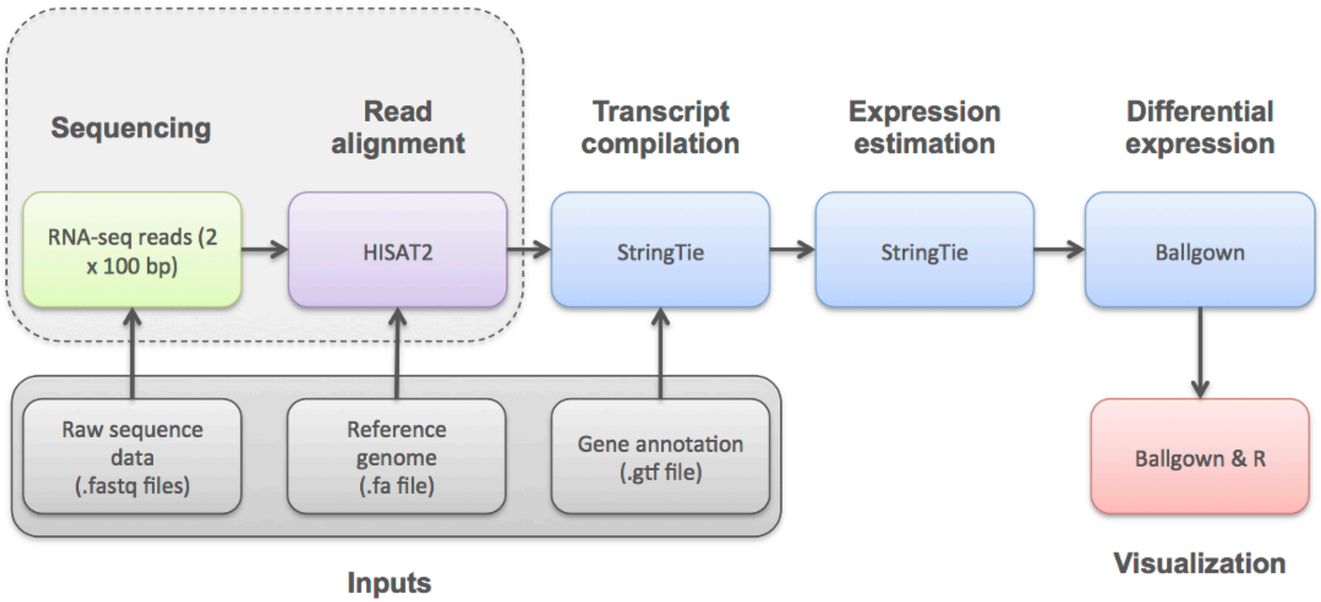
Trinity

- !!! Always have a back-up copy of your raw data !!!
- Would not recommend on personal laptop/computer without a lot of computational power/memory
- Trinity performs *in silico* normalisation and include Trimmomatic to trim low quality reads
- Includes perl scripts to assess the quality of the assembly
  - counts of transcripts
  - N50
  - average contig length
  - total assembled bases
- I would also advise computing BUSCO scores to explore the ‘completeness’ of the assembly according to the conserved ortholog content

Alignment-based transcriptomics

HISAT2

- HISAT2 is a fast alignment program that is capable of aligning both DNA and RNA to a single reference genome
- Splice aware aligner
  - good quality aligner for transcriptomics
  - superseded TOPHAT



Alignment-based transcriptomics

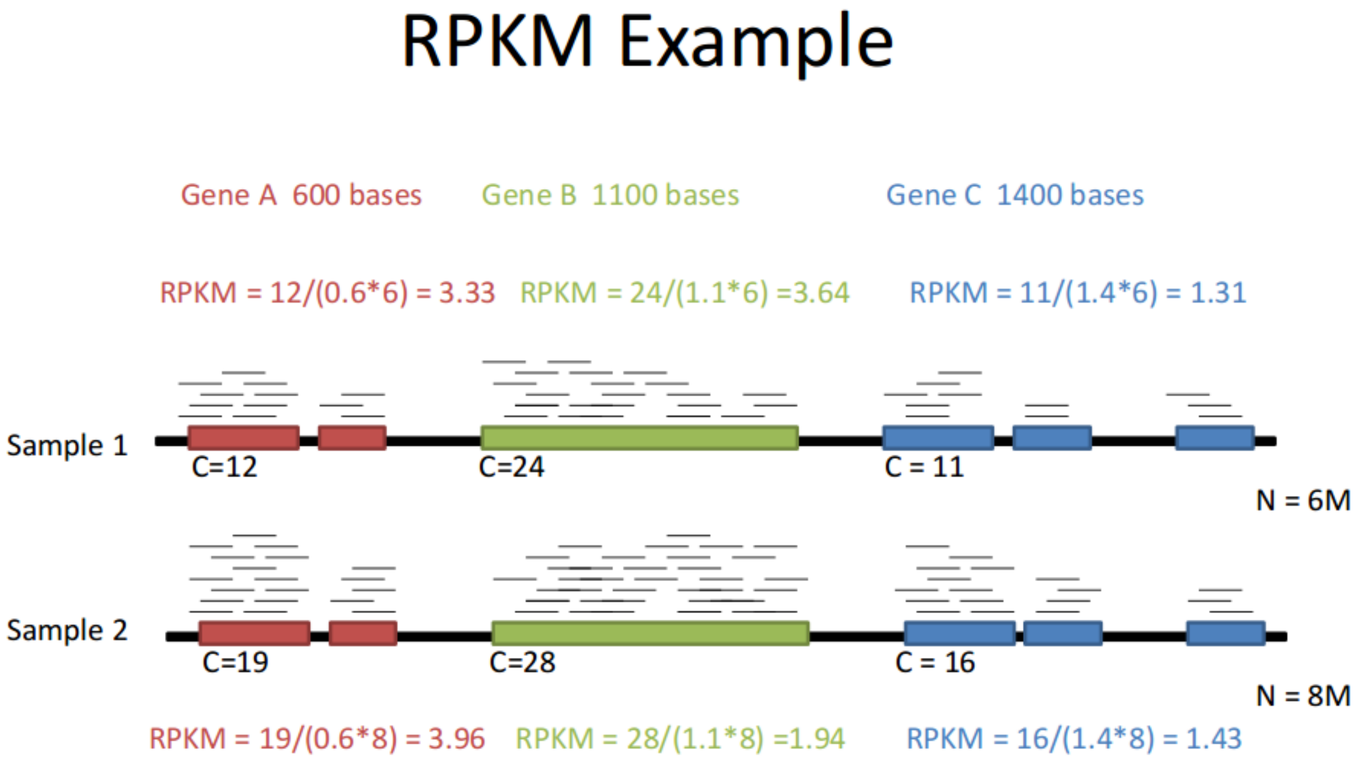
Preparation

- Quality check RNAseq FASTQ files with FastQC
- Index reference genome
  - extract splice sites and exons using gtf associated with the reference genome (usually available on GenBank)
- gtf

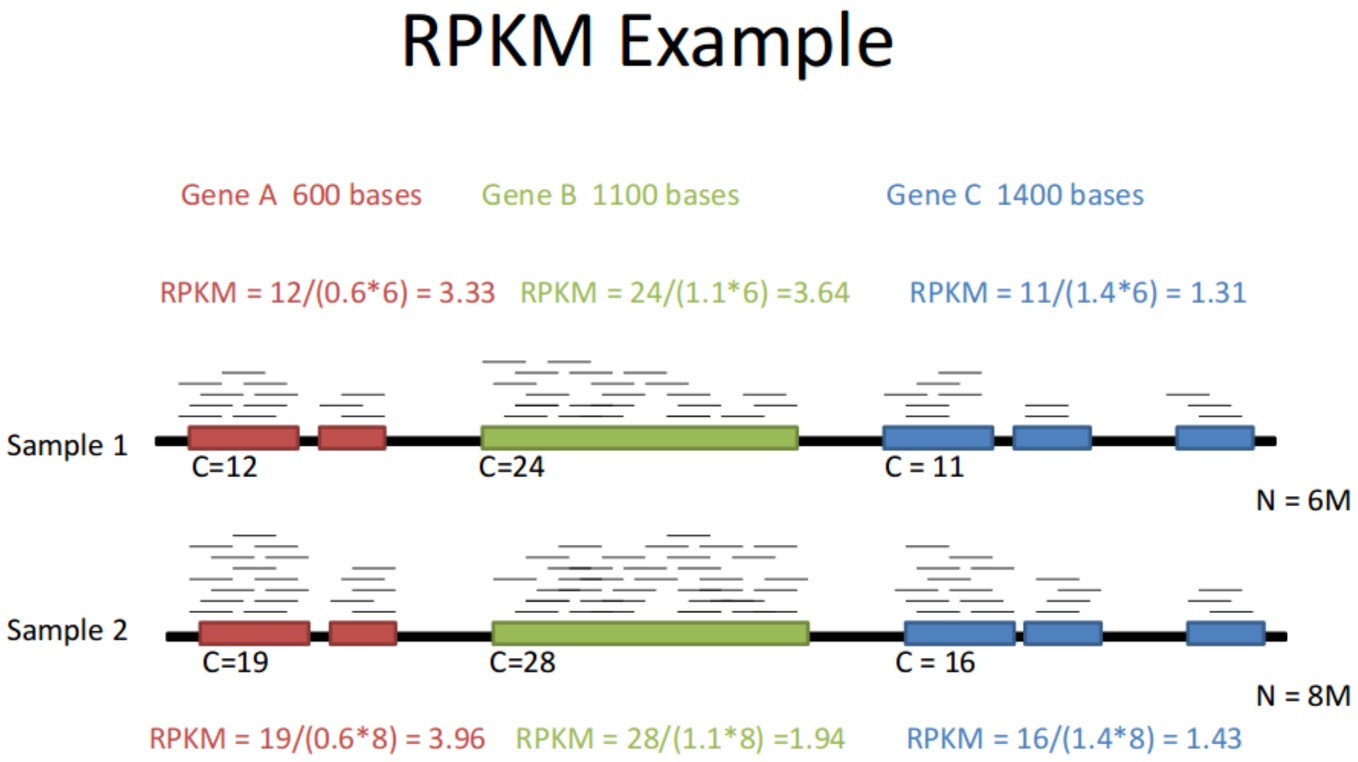
Estimating expression

RPKM, TPM, normalised expression

- RPKM/FPKM - reads/fragments per kilo base of transcript per million mapped
  - downside: fragments biases towards larger genes and number of fragments is related to total library depth
  - upside: attempts to normalise for gene size and library depth



Estimating expression

RPKM, TPM, normalised expression

- TPM - transcripts per million
  - downside: does not account total number of mapped reads
  - upside: considers gene length, alternative to FPKM
- Normalised counts - most basic/ easiest
- No one estimate is better than another